

ZANZIBAR, Tanzania, United Republic of

WMO: 638700

Lat: 6.222S

Lon: 39.225E

Elev: 17

StdP: 101.13

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.4	21.1	14.7	10.4	24.6	16.5	11.8	26.1	10.4	27.9	10.0	28.0	0.2	130	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.8	33.9	26.4	33.1	26.0	32.4	25.8	27.6	31.6	27.1	30.9	26.8	30.4	5.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	29.7	26.1	21.6	29.5	25.9	21.3	29.2	87.7	32.1	85.8	31.4	84.4	30.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.5	8.0	DB	17.7	36.4	2.3	1.7	16.0	37.6	14.7	38.6	13.3	39.5	11.6	40.8	(4)
(5)			WB	16.4	28.7	2.0	0.7	14.9	29.2	13.7	29.6	12.6	30.0	11.1	30.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.2	28.7	28.7	28.6	27.7	26.8	26.3	25.8	25.7	26.1	26.7	27.1	28.1	(6)	
	DBStd	1.48	1.03	1.17	1.22	1.21	1.00	0.91	0.80	0.89	0.82	0.94	0.92	0.97	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6273	581	522	577	531	521	489	490	486	482	519	513	563	(10)	
	CDD18.3	3232	323	289	319	281	263	239	232	227	232	261	263	304	(11)	
	CDH23.3	30834	3674	3324	3617	2835	2302	1849	1544	1580	1822	2356	2587	3345	(12)	
	CDH26.7	9809	1436	1353	1416	819	483	360	283	357	512	747	824	1220	(13)	
(14) Wind	WSAvg	3.5	3.8	3.3	2.6	3.1	3.9	4.2	4.5	3.7	3.7	3.5	2.5	2.8	(14)	
(15) Precipitation	PrecAvg	1557	72	59	191	333	245	64	42	48	54	101	181	152	(15)	
	PrecMax	2160	249	219	417	602	497	211	81	133	156	491	366	303	(16)	
	PrecMin	824	0	2	55	51	19	6	15	12	13	5	47	13	(17)	
	PrecStd	315	57	59	84	160	143	52	23	31	36	99	74	90	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	34.9	35.1	33.1	31.2	30.5	30.1	30.6	31.2	32.3	32.3	33.2	(19)	
		MCWB	26.3	26.7	27.0	26.5	25.5	24.2	23.2	22.9	23.0	24.1	25.8	26.0	(20)	
	2%	DB	33.2	33.8	34.0	32.0	30.4	30.0	29.4	30.0	30.8	31.7	31.8	32.5	(21)	
		MCWB	25.7	26.1	26.8	26.4	25.2	24.0	23.0	22.9	23.0	24.2	25.6	25.8	(22)	
	5%	DB	32.8	32.9	33.0	31.1	30.0	29.2	29.0	29.2	30.1	31.0	31.0	32.0	(23)	
		MCWB	25.6	25.9	26.4	26.3	25.2	23.8	22.9	22.8	23.0	24.3	25.7	25.9	(24)	
	10%	DB	32.0	32.1	32.0	30.4	29.2	28.8	28.2	28.9	29.5	30.1	30.2	31.2	(25)	
		MCWB	25.6	25.7	26.2	26.2	25.0	23.8	22.8	22.7	23.0	24.4	25.6	26.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	28.6	28.0	26.7	25.8	24.9	24.7	25.2	26.5	27.2	27.7	(27)	
		MCDB	32.1	32.6	33.4	31.3	29.4	28.6	28.2	28.3	28.6	29.6	30.2	31.3	(28)	
	2%	WB	27.0	27.1	27.6	27.2	26.3	25.2	24.2	24.2	24.7	25.9	26.7	27.1	(29)	
		MCDB	31.2	31.8	31.9	30.3	29.0	28.2	27.5	27.7	28.3	29.3	29.9	30.6	(30)	
	5%	WB	26.6	26.6	27.1	26.8	25.9	24.8	23.9	23.7	24.2	25.4	26.3	26.7	(31)	
		MCDB	30.5	31.1	30.9	29.7	28.7	27.7	27.1	27.2	27.8	28.8	29.4	30.1	(32)	
	10%	WB	26.2	26.2	26.6	26.4	25.6	24.5	23.5	23.9	23.9	25.0	26.0	26.3	(33)	
		MCDB	29.9	30.5	30.2	29.2	28.1	27.3	26.7	26.8	27.5	28.4	29.1	29.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.8	7.2	5.5	5.1	5.7	6.2	7.2	8.1	7.9	7.0	7.0	(35)	
		MCDBR	7.9	8.5	8.6	6.5	5.9	6.1	6.6	7.6	8.7	8.8	7.9	7.5	(36)	
	5% WB	MCWBR	2.5	3.0	3.1	2.8	2.8	2.6	2.6	2.9	3.2	3.2	3.3	2.9	(37)	
		MCWBR	7.4	8.1	7.4	6.0	5.5	5.9	6.4	7.3	8.0	7.7	7.0	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.482	0.489	0.436	0.406	0.410	0.409	0.415	0.413	0.410	0.410	0.410	0.446	(40)	
		taud	2.259	2.246	2.437	2.540	2.506	2.477	2.425	2.424	2.428	2.445	2.478	2.391	(41)	
	Ebn at Noon		862	856	889	891	860	845	846	872	900	915	922	891	(42)	
		Edh at Noon	145	148	120	104	103	104	111	115	119	119	116	126	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.01	6.19	5.75	4.82	4.50	4.65	4.77	5.12	5.71	6.07	6.09	6.05	(44)	
	RadStd		0.38	0.34	0.38	0.43	0.44	0.19	0.18	0.17	0.20	0.23	0.29	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page